

Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI

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Abstract

Avoiding over and under fitted analyses (OF, UF) and models is critical for ensuring as high generalization performance as possible and is of profound importance for the success of ML/AI modeling. In modern ML/AI practice OF/UF are typically interacting with error estimator procedures and model selection, as well as with sampling and reporting biases and thus need be considered together in context. The more general situations of over confidence (OC) about models and/or under-performing (UP) models can occur in many subtle and not so subtle ways especially in the presence of high-dimensional data, modest or small sample sizes, powerful learners and imperfect data designs. Because over/under confidence about models are closely related to model complexity, model selection, error estimation and sampling (as part of data design) we connect these concepts with the material of chapters “An Appraisal and Operating Characteristics of Major ML Methods Applicable in Healthcare and Health Science,” “Data Design,” and “Evaluation”. These concepts are also closely related to statistical significance and scientific reproducibility. We examine several common scenarios where over confidence in model performance and/or model under performance occur as well as detailed practices for preventing, testing and correcting them.

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The Critical Importance of Over Fitting and Under Fitting

As important over and underfitting are, they are commonly hard to grasp by non-technical users of ML/AI models and systems, not because their basic definitions are hard to comprehend and recall, but because they tend to be created or to manifest in subtle ways and often can be hard to detect before they create significant errors at the time of model application or testing on human subjects. There are also significant and pervasive misconceptions about OF and UF, stemming from earlier stages in the science of ML/AI which we will clarify.

We first present a few classical and pedagogical examples of OF and UF before proceeding with more precise definitions and systematic ways to address them.

Introductory (Pedagogical) Example 1: Who Is the Teacher?

Consider the following thought experiment involving a university course on ML/AI taught by the two lead authors of the present book. The students are tasked with creating a computable rule (i.e. a decision model that can run in a computer) that can classify individuals in the classroom as belonging to either of 2 classes: (a) students, (b) teachers. Numerous variables have been captured including the structural, behavioral and other characteristics of all individuals involved. Some of them are shown in the table below:

Name	Wears suit and tie	Degrees	Presents material to the class	Has beard	Has accent	Wears glasses	Gender	Hands out assignments	Judo black belt	Listed in U catalogue as course instructor	TRUE STATUS (response variable)
Simon	No	PhD	Yes	No	Yes	No	M	Yes	Yes	Yes	Teacher
Aliferis	Some- times	MD, PhD	Yes	Yes	Yes	Yes	M	Yes	No	Yes	Teacher
Smith	No	BSc	Yes	No	No	No	F	No	No	No	Student
Zheng	No	RN	Yes	No	Yes	Yes	F	No	No	No	Student
Singh	Yes	PharmD	Yes	Yes	No	Yes	M	No	No	No	Student
LaFleur	No	BSc	Yes	No	No	Yes	M	No	No	No	Student
Bickman	No	MD	Yes	No	No	No	F	No	No	No	Student
Papado- poulos	No	BSc	Yes	Yes	Yes	No	M	No	No	No	Student
Chang	No	MD, MS	Yes	No	No	Yes	F	No	No	Yes	Guest lecturer
Jones	No	PhD	Yes	No	No	No	NB	Yes	No	No	TA
Schwartz	No	BSc	No	No	No	Yes	F	No	No	No	Auditing student

In this thought experiment there is a number of models that would classify correctly the participants in the class with respect to whether they belong in the teacher or learner class. Some examples of perfectly accurate models to identify teachers are:

- Model 1: hands out assignments and has accent
- Model 2: has PhD and is male
- Model 3: wears suit & tie sometimes OR has judo black belt
- Model 4: name ends in “on” or starts with “a”

The more variables we measure, the more such accurate models we can construct. Such models can achieve 100% accuracy in this class, however *they obviously do not generalize well to other similar ML/AI classes across many other universities. We say that these models are overfitted.*

Consider now the following examples:

- Model 6: Has beard and wears glasses
- Model 7: Has accent
- Model 8: Has beard or a judo black belt

These models can achieve modest/poor accuracy in this class, however they may generalize to similar (low) accuracy to other similar ML/AI classes across the many universities and alternative models exist with better generalizable accuracy. **We say that these models are underfitted.**

Finally consider the following examples:

Model 9: person is listed in the university catalogue as instructor for the course and hands out assignments.

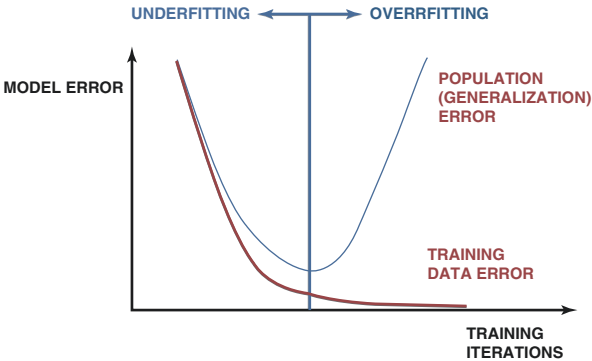
This model can achieve 100% accuracy in this class and will generalize to similar extent to other ML/AI classes across the many universities. **We say that such models are neither under nor overfitted.**

The intuitions gained are that with datasets with a large enough number of variables, it is easy enough to come up with models that are very accurate in the train (discovery) data but that will fail to generalize to the general population. Also that performance in the training data is a poor indicator of performance in the population.

Introductory Example 2: Over and Undertraining ANNs

Figure 1 shows a classical experiment in training ANNs [1]. As the number of weight update iterations increases, the model learns the training data really well and its generalization performance also increases (i.e., error decreases). There is however a “breaking point” (or inflection point) where more training does increase accuracy in the training data but decreases generalization performance.

Fig. 1 Illustration of overfitting and underfitting

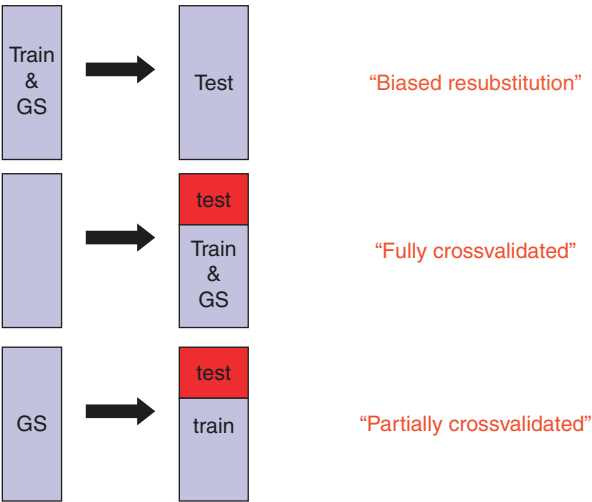


The models to the left of the optimal point are underfitted and to the right are overfitted.

The intuition gained is that there is a level of model “fit” that is ideal for this data and anything above or below that point will lead to worse generalization performance. Performance in the training data remains a poor indicator of performance in the population, however.

Introductory Example 3: Rich Simon’s OF Demonstration for Genomics-Driven Discovery

In bioinformatics where dimensionalities are typically very high and sample sizes small, OF is a particularly important danger. Simon et al published the following empirical demonstration showcasing that depending on how gene selection and model error estimation are conducted, there are different degrees of biasing the estimates of model generalization error [2].



Simon et al. generated models using feature (gene) selection (GS) procedures in data constructed so that there is no predictive signal. They examined 3 protocols for combining the same feature selection and classification algorithms:

Protocol 1 “Biased resubstitution” is when the gene selection takes place on all data and the error estimation also takes place on all data.

Protocol 2 “Full cross validation” is when feature selection is done on a training portion of the data, model is fitted in the training portion and error is estimated in a separate testing portion.

Protocol 3 “Partial cross-validation” conducts feature selection on all data, then models are built in a training portion and model error is estimated in a separate testing portion.

Unbiased error estimation should indicate that a model fit in such data should have no signal, that is perform as well as random coin flipping. Indeed the “fully cross validated” protocol 2 (which we referred to as **nested cross validation** chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”) is indeed unbiased. No cross validation (protocol 1) has large bias that can reach estimates of perfect classification if enough variables are used. Partial cross validation has *in this analysis setting* intermediate bias, smaller than nested cross validation and lower than no cross validation.

The intuitions gained are that we can produce highly biased error estimates if our protocols are not set up properly to avoid bias, especially in high dimensional data. Also **the same algorithms for classification, feature selection etc. can lead to dramatically different quality of results depending of how they are arranged in modeling protocols.**

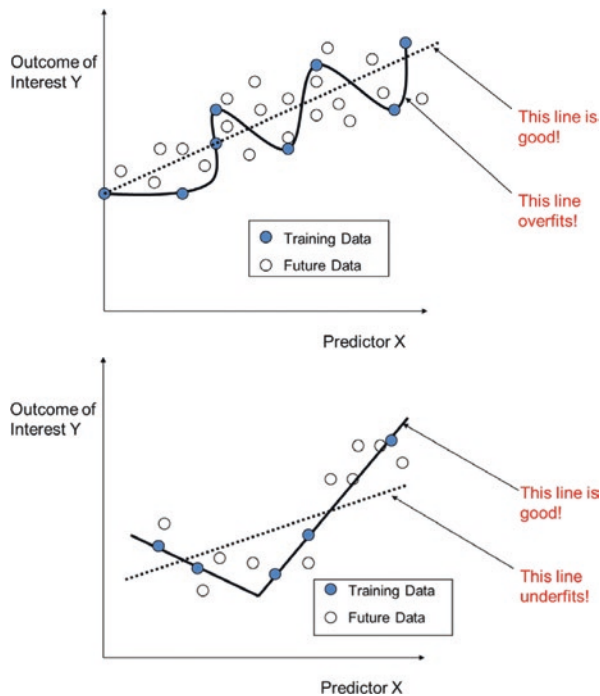
Introductory Example 4: Simple and Complex Surfaces

A classic demonstration of under and overfitting of a regression function of a continuous outcome Y given a continuous input X is given in Fig. 2. We see both training data (blue circles) and unsampled population data (white circles). As shown the complex model (wiggly line) fits the training data perfectly but fails in the population (future) data. A simpler model (straight line) does much worse in the training data, but better in the future data [3].

In Fig. 2 above, the better model is one that is more complicated than the straight line and less complicated than the complex wiggly one.

The intuition gained is that over/under-fitting are directly related to the complexity of the decision surface and how well the training data is fit. Successful data analysis methods balance training data fit with complexity since: too complex model (to fit training data well) leads to overfitting (i.e., model does not generalize) whereas: too simplistic models (to avoid overfitting) lead to underfitting (will generalize but the fit to both the training and future data will be low and predictive performance small).

Fig. 2 Further illustration of overfitting (top, 2a) and underfitting (bottom, 2b)



Definitions of OF and UF; the Broader Pitfalls of Overconfidence and of Under-Performing Models. Bias-Variance Error Decomposition View

Training data error of a model M is the error of M on the training data used to derive M .

True generalization error of a model M is the error of M on the population or distribution, from which training data used to derive M , were sampled from.

Estimated generalization error of a model M is the estimated error (via an *error estimator* procedure applied on data samples) of M on the population or distribution from which training data used to derive M were sampled from.

In controlled conditions, for example in simulation experiments, the true generalization error of any model can be known. Typically the true generalization error is unknown when dealing in non-trivial real-world problems, not constructed in the lab, however.

Typically (i.e., unless training sample is enormous), model training data error used to estimate true generalization error leads to downward-biased (unduly optimistic) estimates of generalization error.

Overfitting a model to data is creating a model that (a) accurately represents the training data, but (b) fails to generalize well to new data sampled from the same distribution (because some of the learned patterns in the training data are not representative of the population).

Alternatively, an overfitted model is often defined as a model that is more complex than the ideal model for the data and problem at hand.

Finally some authors define overfitting as learning “noise” in the data, that is learning idiosyncrasies of the training data that are not present in the population [1].

Similarly, the notion of OF applies at the method, modeling protocol and system level whereby an overfitting ML/AI method, modeling system, or modeling protocol/data science stack have a propensity to overfit models to the data [4, 5].

An overfitting ML/AI method, system, stack, or protocol is one with the propensity to generate models that overfit.

Conversely:

Underfitting a model to data is creating a model that represents the training data sub-optimally and also fails to perform well in the general population. More broadly, an underfitted model will have true generalization error that is larger than the true generalization error of the best possible model that can be fit with the data in hand.

An underfitting ML/AI method, system, stack, or protocol is one with the propensity to generate models that underfit.

A model M cannot be both over and under fitted: the model, either describes the training data well and generalizes worse, or it describes the training data poorly and also generalizes poorly, or finally, it is *ideally fitted* and describes both the training and population data as well as possible. From a complexity perspective, a model is either more or less or equally complex than ideal for the data and problem at hand. We will now delve deeper into these concepts using BVDE (i.e., bias-variance decomposition of a model’s error).

Bias-variance decomposition perspective on OF and UF. As detailed in chapter “Foundations and Properties of AI/ML Systems”, the BVDE describes that a model’s error is (excluding of course measurement noise and inherent stochasticity in the data generating function) a combination of two error components: (a) a “bias” component and (b) a “variance” component. Everything else being equal, a highly biased model is less complex, while a lower bias one is more complex than ideal. Small sample size corresponds to higher variance, while larger sample size, leads to smaller variance error component. *For a fixed sample size and data generating function, there is an optimal model complexity leading to smallest model error possible.*

Model complexity above this ideal level corresponds to over-fitted models, while smaller complexity corresponds to under-fitted models. Moreover, as a model is increasingly overfitted because of increasing model complexity, the fit of the training data will improve and thus the error on the training data will decrease while the true model generalization error in the population will increase.

We further point out a fundamental asymmetry between UF and OF: empirical proof of OF of a model M1 can be obtained by showing that the true generalization error of M1 is higher than what is expected by the performance in training data.

However empirical proof of UF of a model M1 requires showing that there is at least one other model M2 that has better true generalization error than M1. M2 does not need be the optimal model attainable. M1 will also have larger generalization error than the optimal model M_{opt} .

In other words UF is a relative property with respect to some optimal model (or other higher-performing models) achievable under the circumstances and not an intrinsic property of a model under examination. Therefore establishing or preventing UF is harder and more open-ended than establishing or preventing OF.

Whereas at face value establishing that a model is OF and/or UF requires calculating its true generalization error, in practice we can circumvent this (a priori formidable) difficulty by application of unbiased and efficient estimators of true generalization error (see chapters “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” and “Evaluation”). It is also possible to apply statistical theory to infer that a model’s estimated generalization error is not accurate or to infer with high confidence from small-sample estimates that true generalization errors E1 and E2 of models M1 and M2 are not the same (see details in section “How ML/AI Model OC is Generated in Common Practice” below).

We now address two concepts of broader significance:

Over confidence in a model (OC) occurs when the analyst’s estimated generalization (population) error of the model is smaller than the true generalization error.

Under confidence in a model (UC) occurs when the analysts’ estimated generalization error of the model is higher than the true generalization error.

Over performance of a model (OP) (relative to a lower estimated performance expectation) occurs when the estimated population error of the model is smaller than the true population error. It is thus obvious that **Under confidence in a model = Over performance of the same model.**

Under performance of a model (UP) occurs when the true population error of the model is lower than the best possible model, or other better performing models (that can be estimated from same sample size, on average). We can also talk about under performance relative to a high performance *goal* set during the model development planning stages (chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”).

Under confidence in a model (or equivalently over performance of a model) is not considered in practice a serious concern, however it entails opportunity costs that may be significant. For example, it may trigger expensive and time consuming but unnecessary additional modeling efforts, and may deprive productive use of the model (in healthcare or health science discovery) in the meanwhile.

Under fitting is a special case error of under-performance relative to the best/better models that could be produced under the circumstances specifically due to lower than ideal model complexity and so that the lower performance is reflected both in the training data and in the population.

Over fitting may or may not be a case of over-confidence in the models produced (and over- interpretation of the modeling results). For example, overfitting is a special case of overconfidence when the (low) error in the training data is misinterpreted as an indicator of the (higher) generalization error. If however, we overfit a model to the training data but use an unbiased error estimator, then we will not have either over or under confidence in this over fitted model.

The major (high-level) pitfalls that the present chapter addresses is:

Pitfall 10.1

Producing models in which we have over-confidence.

Pitfall 10.2

Producing models that under perform.

With corresponding high-level best practices:

Best Practice 10.1

Deploy procedures that prevent, diagnose and remedy errors of overconfidence in, or overfitting of models.

Best Practice 10.2

Deploy procedures that prevent, diagnose and remedy errors of model under-performance or underfitting.

Before embarking into specific situations where OC occurs, and corresponding remedies, we introduce fundamental general principles spanning statistics and ML that underlie these phenomena. These have value for understanding and developing general approaches to prevent OC and UP.

Fundamental ML Insights About the Three Sources of Overconfidence (OC), Under Performance (UP) and their Relationship to OF and UF: Biased Data Design, Biased Error Estimation, Poor Model Selection

- (a) As discussed, OF in one of its classic definitions occurs when the (low) error in the training data is misinterpreted as an indicator of the (higher) generalization error. This creates a seriously dangerous OC about the model in question. Similarly, if UF means failing to model the training data correctly in both training data and the population and such error is readily available to calculate, why do we ever under fit models?

The answer related to real-life practice is that circa 2023 no professional practitioner of ML uses raw training data error to estimate generalization error. To the extent that even sophisticated but biased error estimators are used, the problems of OF and UF can still occur. **Fundamental insight:** as we explained previously, there is an ideal complexity for a model for which, per BVDE, generalization error is minimized. Unfortunately this complexity is rarely known a priori, hence we typically use empirical data analysis procedures to find the right complexity. These procedures are in practice a combination of search over a space of possible models (i.e., *model selection*) combined with an *error estimation* procedure used to evaluate the merit candidate models examined. If the error estimator is downward-biased (optimistic) and/or the model selection is incomplete, then it is in practice possible to find models that appear as better (=more promising) than they are by themselves (hence OF) or compared to alternative models (hence UF). In either case we encounter OC and/or UP problems.

- (b) **Insight:** Another way to view OC due to OF is that of learning “statistical noise”, or stated differently, over fitting by learning idiosyncratic characteristics or complex patterns of the data sample(s) that are by definition not representative of the population. Even when the error estimation procedures are not naïve (e.g. use of training data error to estimate population generalization error), the total sample itself (comprising both training and validation datasets) is not representative. This is an instance of the error due to high variance per the BVDE.
- (c) **Insight:** alternatively, learning such idiosyncratic characteristics may be the result of actively selecting (as opposed to randomly sampling) training data or because of poor data design such that the training data is not representative of the targeted population and application goals (non- random sampling or more generally mismatch of the available population with the target population as explained in the chapter on data design).

These situations (i.e., poor sampling methods, poor error estimators) can lead to generalization error estimates that are too optimistic (biased downward) and learning spurious patterns translating into error in future application of the model that are not present in the training data.

- (d) **Insight:** with regards to UP, poor choice of training data can lead to UP, and the same holds true for error estimator biases. In addition, poor data design and model selection deficiencies can create UP.

To summarize, from a ML lens, OC and UP originate in 3 different stages/aspects of ML modeling, *and their combination*:

1. **OC/UP created due to data design**, primarily sampling, so that the models and generalization error estimates lead to OC or UP.
2. **OC/UP created due to error estimation** so that the models' generalization error estimates lead to OC or UP.
3. **OC/UP created due to poor choice of model**, that is the choice of model family, model fitting algorithm, model selection procedure etc., are such that models with OC/UP characteristics ensue.

Additional Insights Spanning ML and Statistics as they Relate to OC and UP: Reproducibility, Cross Validation, Nesting and Biased Post-hoc Reporting

- (a) **Insight:** *reproducibility suggests generalizability; operationalizing reproducibility via independent data validation.* A commonly-followed principle of biomedical science is that credible results (e.g., in our context, models that have small generalizable error) have to successfully reproduce, that is, models must have same performance in data independent of the data used to discover these results.

We emphasize that as widely accepted this principle may be, its merits *are not immutable but hold under assumptions*. For example, in a classical statistical context, if we wish to verify that a statistical association found in data D1 reproduces in data D2, everything else being equal, sufficiently high power (type II error probability, of false negative results) and sufficiently low alpha (type I error probability, of false positive results) must be in place if we wish to conclude with high certainty that a reproducible result is a true one. Similarly, if we wish to verify that a model built from data D1 and having estimated generalizable error in the population of E1, by applying on data D2 with estimated error $E2 = E1$, then D1 and D2 must be sampled randomly from the same population and have high enough sample size so that comparison of E1 and E2 is sufficiently powered at low alpha levels.

Without these assumptions holding, an application of the principle of reproducibility may generate both false positive and false negative validation of original results.

The way we typically **operationalize this principle of validity via reproducibility**, is by **well-designed independent data set validation**, and very

commonly, holdout or cross validation varieties (although many other variants exist, the general principles still apply).

- (b) **Insight:** *single dataset independent (holdout) validation is an unbiased estimator and protects against p-hacking and HARKing.* We can use data D1 for fitting a model M1 and then apply it on independent data D2 and compare generalizable error estimates. *This is mathematically equivalent* to randomly splitting the original data into a training (TR) dataset and the rest used for validation or testing (TE) dataset. This is the “holdout estimator” [3] and the error measurement in TE is an unbiased generalizable error estimate.

Application of the holdout estimator protects against so-called HARKing which stands for “Hypothesis testing After Results are Known” [6]. This is because the *effective alpha* of a null hypothesis rejected by both discovery (TR) and validation (TR) datasets is: (*nominal alpha employed in TR * nominal alpha employed in TE*). The total expected false positives in the discovery dataset TR will be (*nominal alpha in TR * number of hypotheses tested*). This is the alpha of using just one dataset. The total expected false positives of original discovery followed by independent validation will be (*nominal alpha in TR * nominal alpha employed in TE * number of hypotheses tested*). In other words the independent data testing reduces false positives by alpha (which is a very small number, typically never to exceed 5%).

For our purposes of modeling with ML/AI, the “hypotheses” in question are typically one or more model(s) for which we test whether its estimated error meets or exceeds a performance threshold.

If more than one validation datasets are used in sequential steps of independent validation, the probability of false findings drops exponentially fast to the number of validations. The effective alpha of being selected in both TR and (k-1) TE datasets is α^k which is a very small probability. For example, if $\alpha = 0.05$ then the effective alpha of reproduced null hypothesis rejection in two independent datasets sequentially is 0.000125. In other words, when more than one validation datasets/steps are used, the probability of false findings drops exponentially fast to the number of validations. These same principles apply to ML/AI models whose validity we want to test.

- (c) **Insight:** *multi-split data validation (NCV) is also unbiased and less susceptible to sampling variation.* Because the split of the original data into TR and TE is subject to random sampling variation and some splits will not be such that both TR and TE are representative of the population (even though TR + TE may be) we often use n-fold cross validation (NFCV) which is also a (in practical terms) unbiased estimator and less susceptible to bad random splits (see chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”). NFCV and its cousin, the repeated NFCV (RNFCV) approximates the less-variant but computationally very expensive (and almost never used in practice) *all splits cross validation* where all splits are employed and averaged over.
- (d) **Insight:** *NCV and NFCV can still be biased via information contamination unless nesting is employed.* Any procedure that transfers information about TE to TR can

“contaminate” the unbiasedness of the CV estimator and lead to bias. Recall the seminal paper of Rich Simon et al. discussed earlier in this chapter, where they showed that feature selection can contaminate (i.e. bias) CV-derived error estimates if it is conducted in all (TR + TE) data and that when it is conducted separately in TR and TE, such contamination does not happen. Here is how this contamination takes place: by conducting e.g., feature selection on all data (using a univariate association filtering procedure in this case), we generate false positive features that happen to appear significant in this (TR + TE) dataset. Then we fit models with these false positives in TR. The model works well in TE because the features are specifically selected to work well in all data (i.e., TR + TE). However the model will not work in the population because the features used in it include false positives (in Simon et al. experiment they were all false positives because the data was constructed to be devoid of predictive signal). In other words, the chosen features were within the false positive expectation of the feature selection procedure and thus by definition will not generalize in the population).

Compare the above scenario with conducting feature selection separately in TR and in TE. The false positives of TR will not generalize to TE (because the effective probability of such random success is α^2 instead of α (see “Fundamental ML Insights about the Three Sources of over Confidence (OC), under Performance (UP) and their relationship to OF and UF: Biased Data Design, Biased Error Estimation, Poor Model Selection”, be hence very small, and thus no strong feature selection bias will manifest. With 10,000 irrelevant features at nominal $\alpha = 0.05$ we will obtain 50 false positives features that will lead to biased error estimation. With separate discovery (TR) and validation (TE) stages we will obtain 50 false positive features from TR but only 2.5 (on average) of them will survive the statistical testing in TE. As it turns out 2.5 random features do not have enough capacity to overfit a random distribution of a random outcome conditioned on 10,000 variables and thus the CV error estimate with independent feature selection is unbiased.

Important notes:

- *The exact same type of bias can happen for any data pre-processing or analysis step* including: data normalization, data imputation, any type of feature construction or transformation, *and in general any other data input to a model that is created by processing the full data* because these operations encode information about the data distribution that modeling algorithms can detect (hence biasing error estimates because of information transference about TE to TR).
- *CV contamination bias does not need to be linked to a supervised analysis step*, that is linking the data processing to the values of the response variable we wish to model. Even access to the joint distribution of inputs in TE (i.e., without reference to the response variable’s values in TE) *or to any subset of variables in TE* may lead to models that have biased CV error estimates. For example, we may construct biased models and error estimates using Principal Component Analysis of the data, a procedure that is usually -but falsely- considered “failsafe” from over fitting (see assignment 12).

- Whereas CV contamination may create biased models/error estimates under the above circumstances, whether it will happen and **the degree of bias greatly depends on the safety measures employed by the unit-level (i.e., component) data procedures**. For example, if in the classical Simon et al. experiment the feature selection is tightly controlled for family-wise errors, the bias will be eliminated (see assignment 1 and later chapter “Lessons Learned from Historical Failures, Limitations and Successes of Health AI/ML. Enduring Problems and the Role of Best Practices”). This implies that we can seek to avoid contamination altogether via two general approaches: one is by application of component procedures that anticipate and avoid contamination bias, or by employing non-contaminating protocols, of which nested CV is the paradigmatic example. We can (and often do) also deploy both measures in combination (see assignment 9).
- (e) **Insight:** *CV nesting decouples TE from TR data and eliminates possibility for CV contamination bias.* Recall from chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” that nested CV is CV where we split TR and TE sets as in usual CV and then split further the TR sets into TRTR (“traintrain”) and TRTE (traintest”) subsets (i.e., applying an embedded “inner” loop of CV inside the TR data of the “outer” CV loop). In addition, all data processing (prior to modeling) happens separately in every TRTE dataset. Because this procedure separates data and conducts data pre-processing operations locally inside TR and subsets of it, there is no information transfer (aka contamination) about the modeling from TE to TR. Consequently, the CV contamination bias is eliminated. For a trace of how nested cross validation operates, and high level pseudo code the reader is referred to chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”. For development of the repeated nested balanced n-fold cross validation procedure by extending the hold out estimator and model selection see [3].
- (f) **Insight:** *Holdout and NCV can also be biased via selective or post-hoc reporting. Both nesting and full protocol specification prevent this bias.* In the context of real-life model development it is still possible to introduce bias in HO (hold-out) and CV if we employ selective and post-hoc reporting. This type of problem occurs when many models are generated and the better ones are reported selectively. When this is a conscious decision by the creators of models, it amounts to a fraudulent representation of their procedures and results. As we will see, selective reporting may occur without ill intent, however, when the analysis plans and protocols are not well-controlled and not well-designed to avoid OC.
- (g) **Insight:** *We can detect bias in biased modeling protocols if they are accurately and thoroughly described using special tests (described later in the present chapter). It is **not possible** to diagnose that a modeling protocol is biased if it is not thoroughly or accurately described.*

Equipped with the general principles leading to OC and UP we will next delve into more details of producing OC/UP models. We will also enrich the discussion with pitfalls and corresponding BPs to avoid OC/UP.

The reader is reminded that the specific pitfalls and best practices discussed here are strongly connected and provide additional technical depth and operational (micro) guidance to the overall (macro and meso) strategies and best practices of chapters “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems” (methods development and validation) and “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” (model development and lifecycle).

How ML/AI Model OC Is Generated in Common Practice

Models Are Allowed to Have Inappropriately Large Complexity with Respect to Data Generating Function Complexity and to Available Sample Size

How do we measure model complexity? In older literature (and even today in some cases) this situation, and overfitting more broadly, was misconstrued as having *too many parameters* in the model (with respect to the available sample size, and the fixed complexity of the data generating function). With newer methods, *the number of parameters does not matter* as long as any number of available protective methods are used to reduce the *effective model parameters* and more generally the complexity of the fitted models, all within an appropriate family of learners (i.e., functions that match the form of the data generating function) [3, 4].

The distinction between original parameters and effective parameters can be made clear using the example of SVMs [3]. In this model family the effective parameters (the support vectors, in other words these data points defining the boundaries of each label class) cannot exceed the sample size. Thus we may have for example 1,000,000 variables and sample size $n = 1000$ which automatically restricts the number of effective parameters to at most 1000 (a 3-order of magnitude reduction over the initial data dimensionality in this example). It is also instructive that SVM mathematical bounds on generalization error do not depend on original data dimensionality but on the support vectors [4].

We have mentioned repeatedly so far that in general the complexity of the model (the effective dimensionality of which is a major factor) must be balanced against the complexity of the data generating function and the sample size. In some more detail: (a) For a fixed complexity data generating function, and a fixed sample size there is an optimal model complexity that leads to the model with the best true generalization error. Similarly, (b) for a fixed complexity data generating function, and a fixed model complexity, there is a minimum sufficient sample size that leads to the model with the true generalization error that is within an acceptable distance δ from the optimal error achievable. This follows from the “Bias-variance decomposition of error” discussed earlier.

Ignoring Statistical Uncertainty

Observing that a model has an acceptable point estimate performance does not guarantee generalizable performance since this estimate may be subject to large variation due to small sample size. The 95% (or higher) confidence interval (CI) of the estimate and/or the 95% credible interval (CrI) must also be taken into account in order to establish that the performance estimate will generalize [7, 8].

X% Confidence Interval (CI) of a point estimate P of the performance of model M developed from a fixed sample of size n : the range of values containing $X\%$ of all point estimates when sampling multiple times, developing a new model for each sample and estimating its generalization error (or other performance metric), when the true value of developing a model from this population is P .

X% Credible Interval (CrI) of a point estimate P of the performance of model M developed from sample of size n : the range containing with probability $X\%$ the true value of the generalization error (or other performance metric) of M applied to the population.

The CrI may be viewed—in Bayesian terms—as equivalent in meaning to the credible region of the posterior probability density function of the model error in the population containing $X\%$ of the total density function symmetrically around the point estimate, given the data and prior knowledge. In practice however, the CrI can and is often estimated empirically with various procedures.

A subtler way to fall victim to small sample-induced statistical variability is when the error estimator used has high variance that is not taken into account, or when among various unbiased estimators a high variance one is used instead of the lowest variance one. For example, both the holdout estimator and the n -fold cross validation (NFCV) estimator are unbiased/near-unbiased (respectively) however the holdout has higher variance than the NFCV [3, 9]. This is why over-reliance on “independent study verification: often (and falsely) treated by journals and others as a “gold standard” is a pitfall that needs to be avoided.

Using Biased Estimators or Introducing Bias in Unbiased Ones

When the procedure for estimating generalization error is not unbiased, or when the unbiasedness is compromised by implementation decisions, the estimates will be upward or downward biased. Omitting correction of this bias or applying an inappropriate correction leading to a downward error estimate leads to overconfidence. At least two common situations lead to such OC in practice:

- (a) The first case stems from using uncorrected or poorly corrected bootstrapping which is a biased estimator because it employs resampling with replacement [10]. The classifier model is produced by the learning procedures who see unnatural replicates of the same true cases in the data and thus may appear to perform better than in real life where such replicates do not exist (e.g., in omics data where the combination of high dimensional data inputs are unique for each subject, yet in bootstrapping based analyses they are viewed repeatedly as if they were naturally occurring in the population).
- (b) The second case occurs in temporal data analyses where there is a progressive distribution shift over time. If the application of an *unbiased* estimator such as cross validation does not take into account time-dependent distribution changes, then a temporal bias will be introduced into a nominally unbiased estimation procedure (see chapter “Data Design”).

Uncorrected Multiple Statistical Hypotheses Tests, “Data Dredging”, “Fishing Expeditions”

The problem of multiple uncorrected statistical hypotheses manifests when a researcher (or a ML/AI discovery procedure) conducts not just one but many tests of statistical hypotheses and does not address the *combined effective false positive rate error across the totality of all tests conducted*. For example, consider an algorithm (or researcher, the exact same principles apply) that conducts a test of association between variables V_i and V_j in a data sample S . Such tests can be intermediate steps of more complex algorithms, or be used to compare and evaluate models.

Assume that for the observed level of association, the data sample size and the desired type II error (i.e., probability to reject the null hypothesis when it does not hold, i.e., the “power” of the test), the type I error (i.e., the probability to generate a false positive rejection under the null hypothesis) may be quite small, often set at the level of at most 5%. Now if the researchers or ML/AI algorithm conducts for example 1000 such tests, it will produce $1000 \times 0.05 = 50$ false positive results. Epidemiologists describe the problematic practice of generating such false positives as “data dredging” or “fishing expeditions” [11].

However these terms falsely imply to some non-technical audiences, the wrong idea that whenever a discovery algorithm conducts massive amounts of statistical tests for “unbiased” or “hypothesis free” discovery, there will be unavoidable massive numbers of false positives. This is not true, however, as these multiple testing *can be corrected with many powerful and practical ways* as we will show in section “Preventing, Detecting, and Managing UF/UP”.

The problem of uncorrected multiple hypothesis testing can also manifest when we produce a number of models, the estimates of generalizable error of which have considerable variance due to small sample size (as explained before), and we falsely conclude that one or more have desirable performance because of uncorrected multiple testing of the significance or the performance estimates.

Note that so far it should be clear that the notion of OC is not confined to model building and ML/AI, but is more general since it encapsulates broader notions of producing generalizable knowledge from small sample data, and is also related to classical statistical considerations of reliable estimation and inference.

Selective Reporting of Results, “Filedrawer Bias”, “Publication Bias”

Imagine a modeler that aims to build an outcome classifier model for outcome O on the basis of data inputs V . He proceeds as follows: he develops 100 models by using a variety of techniques and estimates the generalization error for each model. Then he reports the model with the best error estimate but does not report that this was selected out of 100 models. This setup will invariably lead to over fitted/OC model reporting as evidenced by the following simple demonstration: imagine that none of these models has error better than flipping a coin (i.e., by chance, e.g. 50% accuracy for a binary outcome in a distribution with prior probability of positives equal to 50%). However if the 99% CI of estimated error for (e.g., sample size = 50 the sample size used is [0.3, 0.68], then we expect that half of the models will show accuracy >50% and some will be higher than 65%.

The data scientist of this hypothetical experiment may also apply a calculation of 95% CI of a model that performs at accuracy 68% in this sample size level ([0.53, 0.80]) and may conduct a statistical test showing that the produced “best” model is statistically significantly better than random chance (at 5% alpha). All of this string of errors and over-interpretations is *undetectable by statistical tests or by reviewers unless they know the precise selection protocol employed*.

“Analysis Creep” and Uncontrolled Iterative Modeling

A milder, all-too-common and often well-intentioned version of the file drawer bias pitfall occurs in the form of non-rigorous iterative analysis, which is sometimes referred to as bias due to “analysis creep”. We illustrate by extending the previous hypothetical scenario:

Consider the data scientist of section “Selective Reporting of Results, ‘Filedrawer Bias’, ‘Publication Bias’” who this time builds the outcome classifiers with careful

cross validation where the data modeling decisions are fixed in the training data and error estimation takes place in the testing data portions of the cross validation. Assume that estimated error is 20%. The PI of the scientific project (or project manager of a commercial product based on the model) reviews the results and suggests trying out an additional classifier algorithm. The data scientist does so (repeating the same modeling protocol but with the new classifier) and produces a new model with error 17%. An external consultant reviews the results and suggests specific data transforms and feature construction which lead (by application of same analysis protocol on the same data) to a model with error 14%. In yet another “improvement” step a new hire in the data science team decides to explore some recalibration method that leads to a model with estimated 10% error. And so forth, until the final model appears to be a near-perfect one.

The problem with this scenario is that no measures are taken to isolate the introduction of new methods from the statistical variation of error estimation so that *the final model’s error estimates are not over fitted to the specific train-test configuration used*.

An especially challenging aspect of the “**analysis creep**” problem that makes it invisible to even the most rigorous scientists who are not experts in data science, is that each step may be well designed and perfectly appropriate as a 1-step analysis, but when a series of such steps is executed, then over fitting and over confidence take place.

Choice of a Few and Non-representative Datasets; Unusual Populations; Broad Claims From Too Few or Too Easy Datasets

Just like non-rigorous (intentionally, or not) selection of models can create over fitted models and over interpreted results, the same is true for datasets.

- This problem is particularly common in ML/AI method development where the developers of methods often choose themselves the datasets to test their methods. It is entirely possible that developers may choose to report performance of their methods in data that is “friendly” to the methods, and to omit reporting (or even testing) performance on harder datasets.
- This situation is also common in ML and data science competitions that pit dozens or hundreds of methods against one another over one or a small number of datasets such that the results are typically overly-specific to the small choice of datasets (as well to the specific data design and performance metrics used).
- A related problem arises in applied discovery settings where multiple data sets exist and only a subset is used for development and testing.

The above scenarios create a lack of generalizability problem where certain methods or models perform well in some highly selected data or populations and therefore fail to generalize beyond those (chapters “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems” and “Lessons Learned from Historical Failures, Limitations and Successes of AI/ML in Healthcare and the Health Sciences. Enduring Problems, and the Role of BPs” give more details in the context of limitations of competitions).

Many Teams, Same Data, No Coordinated Unbiased Protocol

Yet another version of the “analysis creep” and uncontrolled iterative or parallel modeling pitfalls occurs when (typically) in cooperative consortia or other large scale collaborative efforts, several teams are working on analyzing the same data with different algorithms, analysis protocols, error estimators etc.

In the absence of a coordinated unbiased model selection and error estimation protocol that applies to all analyst teams, the statistical variation of even good modeling methods with same large-sample performance can lead to large apparent differences in performance in the discovery data that do not generalize to the population. The problem can be further compounded when the various teams employ learning algorithms, model selection and error estimation methods with widely different characteristics where the selected model overall can simultaneously suffer from under-performance and over-confidence in it (see chapter “Lessons Learned from Historical Failures, Limitations and Successes of AI/ML in Healthcare and the Health Sciences. Enduring Problems, and the Role of Best Practices” for findings from a major centralized benchmark study revealing related problems).

Hard-to-Reproduce, Non-Standardized Data Input Steps

In some types of clinical as well in discovery data modeling the data inputs may be subjectively assessed and these assessments may have a low degree of reliability across different individuals or settings. If this aspect has not been incorporated in the model performance estimation (e.g., a single observer is responsible for all subjective assessments in both training and test data) then inflated performance estimates are produced.

Examples of this pitfall exist in a variety of settings, for example in assessment by surgeons of operation aspects, in pathologist evaluation of slides for less-than ideally standardized features, skin lesion assessments, proteomic spectra peak determinations, etc.

Normalization/Data Transforms that Require Entirety of Sample

In some modeling settings data needs to be processed (e.g., via normalization, discretization or other operations) by looking across the totality of data, that is including train and test data. This automatically creates the previously-discussed “contamination” of the error estimates of the test set by the train set weakening their independence and creating a bias in the error estimates.

Moreover from a purely practical viewpoint, the application of these models on new data not used in the model development/evaluation cycle is problematic because it requires re-normalization (or re-processing) of *all data* starting from early model development to the latest application.

Learners Learn the Wrong Patterns via Spurious Co-Occurrence

A classical example from epidemiology, involving causality, is the “yellow finger” which when is due to tar-staining from smoking, predicts several diseases resulting from smoking (e.g., lung and cardiovascular diseases, cancer). However the classifiers in these cases cannot discover that eliminating the yellow stains does not alter the disease risk. That requires eliminating the confounding cause (smoking) and thus such purely predictive findings/models do not generalize when interventions are considered.

Selective Control of Factors that Can Lead to OC

Recall that R. Simon et al. showed the role of feature selection for error estimation bias in high dimensional disease classification in “complete”, “partial” and “no” cross validation. *Every single analysis step and parameter (not just feature selection) that affects the performance of models and the error estimators has to be controlled accordingly.* This includes hyperparameters, model families, normalization discretization, imputation etc.

OF and OC Problems in Patient-Specific Modeling

In recent years efforts have been made to develop models specifically tailored to individual patients. These are commonly based on time series data obtained from each individual and they may have predictive or causal foci [12, 13].

The advantages of such modeling is that (1) they-- may avoid masking and distribution-mix effects of population data and (2) may be able to focus more effectively on mechanisms and characteristics of individuals for precision and personalized medicine. Possible disadvantages are that they (3) may require dense time series data, (4) they may fail to leverage vast amounts of data from other individuals that apply to all individuals in the population, (5) by their very nature they cannot model severe and irrevocable, or rare or singular outcomes (e.g., death), (6) do not

deal well with abrupt distribution shifts of the individual (whereas same shifts may be learnable at the population level and anticipated by population models) and (7) do not provide guarantees for generalizing to other individuals.

Characteristics (6) and (7) are therefore related to possible over fitting and over confidence.

Bespoke and Hand-Created AI Models

In certain types of AI modeling, most commonly in mathematical and engineering-based modeling that may incorporate limited data-driven aspects and is not fully automated as in ML, the ability to generate large numbers of models by the same individuals and over short periods of time is severely limited. This precludes the collection of large numbers of models and datasets in which the success of such models can be rigorously statistically evaluated. It is entirely possible under these circumstances for a few models to be performing well in some task yet the overall model building process is neither salable nor demonstrably generalizable. Creation of a handful of successful models by hand and evaluation on a handful of cases, for example a hand-crafted model to predict the safety of a drug in a specific RCT, may say very little about whether this model would apply to other RCTs and even less about whether the modeling methodology could be carried out by other modelers or for other drugs.

How UP ML/AI Models Are Commonly Created

Not Considering the Right Method Family, Not Considering Enough Method Families in Model Selection

A common reason for under performant models is not exploring the right model family in model selection. For example, when the data generating function is non-linear and discontinuous but we explore only linear regression models. When the right family is not known a priori the corresponding problem is not exploring enough method families in model selection.

The above are common occurrences when data scientists have strong preference for a small number or narrow methods, or when vendors focus on a specific technology that is used across diverse tasks even when it is not the most appropriate for the task.

Insufficient Data Preparation

This pitfall applies to all steps typically used for data preparation such as feature construction and selection, normalization, discretization, distribution transforms, etc. Such steps can greatly enhance model performance when employed correctly or hurt performance when they are ignored or conducted sub optimally.

Insufficient Model Selection Hyper Parameter Space

This pitfall occurs when the right model families are explored but without sufficient exploration of their hyper parameter values.

1-Step Modeling Attempts

In chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” we elaborated on the importance of initial modeling which typically must subsequently be refined and enhanced (since the first attempt seldomly meets performance goals in complicated modeling problems). A 1-stage analysis procedure does not benefit from a graduated understanding of the data and task at hand and their interaction with the modeling method deployed.

Ignoring Best Known Methods

In some cases for specific domains and tasks prior theoretical and empirical work has established the predominance or superiority of specific classes of methods. It is therefore likely to under perform if these methods are not included in the model selection, at a minimum as “starting points” with baseline performance that other methods must match or exceed. See chapter “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems” for proper scope of empirically evaluating new methods and for appraising existing methods.

Ignoring Official or Reference Specifications on Methods Use

A commonly-encountered pitfall is applying strong methods but in ways that are inconsistent with suggested use. These suggested uses include: (a) the ways these methods have been previously tested during reference method development and in validation phases leading to established strong results; (b) the specific ways the inventors of these methods have used them in the primary (“official”) publications associated with them. Chapter “Lessons Learned from Historical Failures, Limitations and Successes of AI/ML In Healthcare and the Health Sciences. Enduring Problems, and the Role of Best Practices” describes case studies where such bias led to suboptimal performance.

UP Problems in Models for Individual Patients

As explained previously in “OF and OC Problems in Patient-Specific Modeling”, individual-specific modeling has a priori both advantages and disadvantages over

population-wide modeling. Characteristics (3), (4), (5), (6) and (7) are linked to possible under performance of such models relative to population based models.

Failing to Obtain Power Sample Analysis and More Generally to Control Effects of Sample Size on Modeling

In the absence of a power-sample analysis or knowledge of learning curves, the modeler does not know whether better results can be obtained by increasing the sample size, or what is the minimum sample size needed to reach the stated goals of the modeling.

In ML the power-sample calculation problem is more complicated than traditional statistical power-sample analysis. This is because in inferential statistics we need to know what is the required sample size to reject a null hypothesis with a desired alpha and power, and closed formulas exist that describe this relationship for applicable statistical tests. In ML however, in addition to the need to test a model's performance against a null hypothesis, *we first need to find a good model*. Thus the power-sample calculation is further complicated by the learning protocol's *learning curve* that is the function that describes the generalization error of the best model learned (on average) as a function of sample size. Depending on the problem, learning curves may suggest that we need more or less sample size than the one needed to reject a null hypothesis centered on the model's performance. To make things worse, learning curves are not known a priori for the vast majority of practical problems.

Based on the above we can summarize on the pitfalls related to OC and UP including OF and UF.

Pitfall 10.1

Producing models in which we have over-confidence

10.1.1. Models are allowed to have inappropriately large complexity with respect to data generating function complexity and to available sample size.

10.1.2. Ignoring statistical uncertainty of strong point estimates of performance.

10.1.3. Using biased estimators or introducing bias in unbiased ones.

10.1.4. Not correcting multiple statistical hypotheses tests.

10.1.5. Selectively reporting strongest models/results.

10.1.6. Conducting uncontrolled iterative modeling and succumbing to "analysis creep".

10.1.7. Using non-representative datasets, unusual populations, and making strong claims from too few or too easy datasets.

10.2.8. Not coordinating analysis over many teams and same data via appropriate unbiased protocols designed for collaborative work or competitions.

10.1.9. Using hard-to-reproduce, non-standardized data input steps.

10.1.10. Employing normalization/data transforms that require entirety of sample.

10.1.11. Allowing learners to learn the wrong patterns via spurious co-occurrence; uncontrolled structural relations and biased sampling; and ignoring domain knowledge that reveals the above.

10.1.12. Controlling only some of the factors that can lead to OC.

10.1.13. Inappropriate modelling for individual patients and over interpretation of their generalizability.

10.1.14. Insufficient studies of scalability and generalizability in bespoke hand-created AI models.

Pitfall 10.2

Producing models that are under performing

10.2.1. Not deploying the right model family in model selection; not exploring enough method families in model selection.

10.2.2. Insufficient data preparation.

10.2.3. Insufficient exploration of the hyper parameter space during model selection.

10.2.4. 1-stage modeling attempts.

10.2.5. Ignoring best known methods for task and data at hand (either as baseline comparators or starting point).

10.2.5. Ignoring official specifications and prototypical (reference) use of employed methods.

10.2.6. Models for individual patients: lack of dense time series data, failure to leverage population models that apply to the specific individual (including ignoring or under modeling severe and irrevocable rare or singular outcomes), not addressing well abrupt distribution shifts of the individual (whereas same shifts may be learnable at the population level).

10.2.8. Failing to obtain power sample analysis and more generally control effects of sample size on modeling.

Preventing, Detecting, and Managing OC and OF

We now address specific best practices for preventing overfitted models and over confidence in models.

Manage Model Complexity with Respect to Data Generating Function Complexity and to Available Sample Size

Whereas manually balancing complexity against sample size is a formidable hurdle, well-designed modern ML methods, protocols, and systems encapsulate multiple methods that achieve this balance automatically or semi-automatically:

1. **Regularization:** is a methodology whereas model parameters' values are driven to zero by model fitting algorithms, as much as data allows. Regularization is broadly used by “penalty+loss” learners (see chapter “An Appraisal and Operating Characteristics of Major ML Methods Applicable in Healthcare and Health Science”) for example SVMs, Lasso regression, regularized classical statistical variants such as regularized Cox regression, regularized Logistic regression, regularized Discriminant Function analysis, etc. The “loss” term is a mathematical expression of how accurately a model represents training data, whereas the “penalty” term captures the combined complexity of the model (e.g., sum of squared weights of inputs). Regularization is sometimes closely related to the notion of function smoothness, that is the preference for modeling functions in which the impact of a small change in the data generating function inputs leads to small changes to the classification output (this is mathematically equivalent to the “maximum margin classifier” inductive bias of SVMs).
2. **Dimensionality reduction:** is a set of methods that map the original input variables to a much smaller number of mathematical combinations (see chapter “An Appraisal and Operating Characteristics of Major ML Methods Applicable in Healthcare and Health Science”). A prototypical example is Principal Component Analysis (and variants) where the original input variables are replaced by independent linear combination functions (the principal components, such that the totality of data variance is captured by the totality of the principal components). The fitted models use as inputs the reduced input representation or a subset thereof.
3. **Feature selection:** (see chapters “Foundations and Properties of AI/ML Systems” and “An Appraisal and Operating Characteristics of Major ML Methods Applicable in Healthcare and Health Science”) is a set of methods that select a small number from the original variables such that ideally all information about the response variable is retained and all redundant variables are discarded. Strong feature selection helps reduce model complexity to the absolute necessary (i.e., only those features that have indispensable information about the response).

4. **Bayesian Priors and Bayesian ensembles:** in Bayesian maximum a posteriori model selection-based modeling, prior probabilities over the model space considered can be used to ensure that models with appropriate (smaller) complexity are given more attention in smaller sample sizes and as sample size grows more complex models can be selected. Moreover, via Bayesian Model Averaging, many (all in theory, but just a few in practice in most practical settings) models can be combined to provide an “ensemble” classification. Complex models are expected to have smaller posteriors in small sample sizes than simpler ones and thus to drive more the overall ensemble decisions whereas in larger sample sizes, the opposite is true.
5. **Algorithm-embedded complexity control:** several ML algorithms have embedded means to control complexity of produced models. For example, decision tree learners prune the trees when they reach branching points with small sample sizes. Random forest learners apply feature selection at each branching point of each fitted tree and forbid trees larger than a set size (which is a tunable hyper - parameter). ANNs map large input spaces to potentially smaller hidden layer spaces or incorporate pruning and other regularization steps. SVMs transform non-linearly separable input spaces to linearly separable ones via kernel functions. Structured Risk Minimization in SVMs progressively considers classes of models (corresponding to kernels) of strictly increasing complexity with guarantees strictly monotonic improvements in generalization error. Boosting methods start from simpler models and extend them to address only the cases not classified correctly. And so on (see chapter “An Appraisal and Operating Characteristics of Major ML Methods Applicable in Healthcare and Health Science”).
6. **Statistical model/data complexity measures:** classical examples being the AIC and BIC metrics [7] that are used in statistics to characterize models with respect to their complexity and fit against the training data. Simpler models are preferred by the human analysts, everything else being equal.
7. **Model selection and combination approaches:** it is common to create and apply ML/AI model fitting and selection protocols that combine several of the above approaches.

Using nested cross validation approaches to find the best models and estimate their generalization error is a common approach that allows model complexity to grow only as much it helps improving estimates of the true generalization error.

Characterize and Manage Statistical Uncertainty

Typically this entails: (a) Calculating confidence and credible intervals for models (and of the models’ parameter values when appropriate). (b) Testing models against the null hypothesis (commonly being that: there is no predictive signal in the data at hand, or a network model has properties no different than a random one, etc.). This is easily accomplished with a standard label reshuffling or other randomization tests

(chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”). (c) Conducting tests of model stability (when appropriate) to sample size. (d) Obtaining measures of stability of model output or decisions with respect to variation in the model inputs. (e) Reducing sampling variance by choice of most powerful/low variance estimators and protocols. Especially for n-fold cross validation schemes, repeating the analysis >50 times empirically has been shown to reduce train-test split variance [14].

Use Unbiased Estimators or Correct Estimation Bias

Among classical estimators and model selection protocols Repeated n-fold cross validation (RNFCV) is a particularly robust error estimator which can also be used for powerful model selection when nested (RNNFCV). In larger sample sizes a repeated nested holdout may be a more computationally efficient alternative. Finally, in very small sample situations a repeated nested leave on out can be used as alternative to RNNFCV [3].

We also recommend **“locking” models** at some predefined stage in the modeling process and not allowing further tampering with locked models. Publishing open-box models can certainly provide a strong form of such locking, although it may also be employed as an internal strategy during model development. A similar, very stringent but much less practical best practice is **preregistering ML studies and models** [15].

The exclusive use of independent data testing for establishing or testing for generalizability is commonly and often required by journals, funding study sections etc. It is NOT recommended however as a single, or “privileged” validation methodology in the present volume, since it is subject to between and within-population sampling variation so that discrepancies between the model’s performance in the discovery + testing CV datasets and the independent validation dataset may be due to: (1) sampling from a different population, or (2) not having perfect power in the independent validation dataset (see section “Additional Notes on Strategies and Best Practices for Detection, Analysis and Managing Both OC and UP” for Details). The latter danger is especially salient in domains where sample sizes are never very large, or are very costly, ethically challenging, or slow to obtain. Nested CV by comparison eliminates the first source of errors since we ensure that the discovery and validation datasets come from the same population. See also chapter “Lessons Learned from Historical Failures, Limitations and Successes of AI/ML in Healthcare and the Health Sciences. Enduring Problems, and the Role of Best Practices” for a striking demonstration of variability between discovery and validation sets and results in a landmark benchmark study where same analyses were done with original and swapped discovery-validation sequences [16].

Correct or Control Multiple Statistical Hypotheses Tests

The venerable but outdated Bonferroni correction is not recommended since it reduces the power of the discovery procedure dramatically relative to more modern procedures. The Benjamini- Hochberg (or similar) methods for correlated and uncorrelated p-values can be used to more effectively control the acceptable ratio of false positives (e.g., the analyst can set thresholds on p-values that does not lead to more than 10% false positive rate on the reported results) [17].

We also note that certain algorithms, for example constrained-based causal modeling algorithms (e.g., GLL, LGL and others) have embedded control of false positives due to multiple statistical testing [18].

Thoroughly Specify and Report the Procedure Used to Obtain Results

The entirety of the analyses and modeling applied on data must be reported so that the possibility for over fitting is properly assessed by third parties. We emphasize that even if the original model developers share their modeling algorithms and data in full, unless they specify the analyses employed in their entirety (i.e., entirety of model selection and error estimation steps), *it is not possible to determine whether the models are over fitted without additional verification in independent data*. On the contrary, when the entirety of the analysis protocol is disclosed, both the final model's over fitting and the whole protocol's propensity to over fit or OC can be assessed (often with a simple label reshuffling test as demonstrated in chapter "The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models").

Conduct Iterative and Sequential Modeling via Unbiased Protocols

A robust and proven such protocol is the previously mentioned RNNFCV protocol in which every new method or parameter value added to the previous analysis steps is incorporated in the nested model selection along with all other methods **and the whole modeling is repeated from scratch (ideally with previous model fitting steps cached for improved tractability)**. Other such protocols may be constructed, but it is essential for the modelers to establish first their robustness to bias due to iterative modeling. Note that when we perform model selection over k algorithms and their associated hyper parameter value sets in RNNFCV, the results are mathematically equivalent if we first model-select over the first methods, then insert the

second, then the third and the winner of the first two, and so on until all methods have been examined. This demonstrates that since doing all methods at once is not overfitting/producing over-confident error estimates, the sequential procedure over the same set of methods, will not either.

See assignment 8 for a practical demonstration of proper vs improper iterative modeling.

Use Representative Datasets, Appropriate Populations and Make Generalizability Claims from Appropriate Datasets

In all modeling settings, clinical or biological criteria must be used to establish appropriateness of data used. Rigorous phenotypic definition and extractions from the EHR, for example, will ensure that only and all human subjects that apply to the modeling goals are considered for discovery and validation (see chapter on Data Design).

When new method development or validation and benchmarking are pursued, we recommend using all publicly available datasets that apply to the task and if they are too many, to use a randomly-stratified selection of representative datasets (e.g., with certain distributions, dimensionalities, sample sizes, etc.). See also chapter “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems”.

In situations where discovery is pursued via **secondary analysis of many pre-existing datasets**, it is very common for results to disagree among the various datasets. One way to address the problem is to “**Round-Robin**” **analysis** and examine the nature and robustness of results when some of the data are used for discovery and some for validation.

It is often useful in such situations of multi-dataset analyses, to adopt a different perspective and **focus not on whether model M or property P discovered, e.g., in datasets 1–10 holds in datasets 11–20 but, rather, what are the variant and invariant properties (both predictively and structurally) across this collection of datasets** and what is the **robustness of models** developed in a subset of the datasets on the remainder datasets (with the round-robin analysis conducted so that it examines or approximates all possible discovery-validation splits).

Stated differently, the discrepancies between datasets (and models summarizing properties of these data) should not be viewed automatically as “errors” but also considered as potentially valuable indicators of systematic differences between health care systems, research designs, model organisms etc., depending on the data measurement and sampling designs used.

Coordinate Analysis over Many Teams and Same Data via Appropriate Unbiased Protocols

The recommendations of “Conduct Iterative and Sequential Modeling Via Unbiased Protocols” apply unchanged here as well.

Use Easy-to-Reproduce, Standardized Data Input Steps

Techniques to facilitate this practice include **automating all subjective input measurements** and establishing equivalence or sufficiency of their information content. Alternatively, establishing that subjective measurements can be standardized via protocols that ensure **low interrater variability**.

Employ Normalization/Data Transforms that Do Not Require Entirety of Sample

This is self-explanatory and follows directly from the nature of validation-to-discovery information contamination that biases CV error estimates as explained previously.

Prevent Learners from Learning the Wrong Patterns via Spurious Co-Occurrence; Control Structural Relations and Biased Sampling; and Incorporate Domain Knowledge and Face-Validity Expert Testing that May Reveal Spurious Learning

Essential to the above are robust batch processing bias and error detection and correction protocols such as the ones routinely used in high-throughput omics assay-based studies. Moreover, causal modeling algorithms can reveal spurious and confounded relations and patterns of bias. In addition, a diversity of datasets that fully covers the space of application of the desired models must be used for training and validation. Finally, *model explanation techniques* can be valuable by revealing exactly what the learning algorithms and corresponding models created by those have learned especially when combined with expert review of such models (see chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”).

These latter precautionary practices unfortunately rely on existence of sufficient domain theory that can be used to detect anomalies in the models. It is entirely possible (and indeed common) in certain domains for robust such theory to be lacking, however (e.g., in high-density omics studies, or complex mental health/human behavior and other domains where the complex mechanisms governing the data generating processes have not been conclusively or completely established). It is important in such cases to non over-interpret models and to test the propensity of

human experts to construct invalid conceptual explanations in support of models (see chapters “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems” and “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” for expert biases and modeling of expert judgment). It is equally important to always consider the possibility that models encapsulate valid new knowledge previously unknown in the field, thus an expert’s rejection of some model or result should be a piece of evidence in evaluating model validity and not grounds for immediate dismissal.

Control all Factors that Can Lead to OF/OC via TE→TR Contamination, by Using Nested Model Selection

The most important way to operationalize this is to use a nested protocol for cross validation and error estimation as previously explained, and at the same time ensure that all possible data analysis steps that may transmit information from the test sets to the train sets (and thus introduce bias in the error estimation) are isolated inside the nested part of the protocol.

Carefully Combine Modelling of Individual Patients with Population Modeling

Most importantly, any individualized modeling must be compared with population modeling and ensemble (combined) with population models whenever appropriate and feasible.

Do Not Over-Interpret the Generalizability of Bespoke Hand-Created AI Models Unless Sufficient Validation Data Can be Obtained to Support Such Claims

Unfortunately bespoke, hand-crafted modeling efforts typically cannot - by their very nature - be readily evaluated by automated procedures and data for such evaluations are scarce. It is prudent in such cases to address modeling successes and failures as isolated incidences.

Whenever it is feasible to create automated computable procedures that replicate the bespoke methodologies, this allows transitioning non-scalable human modeling to scalable and testable AI/ML modeling with obvious advantages for increasing the scale, scope, speed, cost-effectiveness and verifiability of similar modeling in other problem domains/settings.

Preventing, Detecting, and Managing UF/UP (Table 1)

Practices 10.5.1.–10.5.7. in Table 1 are self-explanatory and follow directly from the principles presented earlier. The next best practice 10.5.8. require some explanation, however:

Conduct Power Sample Analysis and More Generally Characterize the Effects of Sample Size on Modeling. Use Dynamic Sampling Schemes Whenever Appropriate

With regards to the ML/AI power-sample planning as introduced in “Failing to Obtain Power Sample Analysis and more Generally Control Effects of Sample Size on Modeling”, contrary to classical statistical hypothesis testing where closed formulas exist to calculate the minimum required sample for achieving a desired alpha (% of false positive rejections of the null, or type I error) and beta (probability of false negative failure to reject the null under the alternative hypothesis, or type II error, aka power) levels for some statistical test of choice, when designing ML/AI modeling, two additional factors come into play the first relevant to predictive modeling, and the second related to causal modeling:

- (a) The **learning curve** of the used learning algorithm. The learning curve describes the errors of the algorithm’s output as a function of sample size. Generally the learning curves are not known and closed formulas do not exist.

Table 1 Lists specific practices for preventing under fitted models

10.5.1. Deploy and explore all appropriate learning method families in model selection
10.5.2. Deploy and explore all relevant data preparation steps to the domain and task at hand
10.5.3. Systematically and sufficiently explore the hyper parameter space
10.5.4. Anticipate several preliminary and refinement modeling stages and incorporate in nested designs to avoid overfitting
10.5.5. Inform analyses by literature so that best known methods for task and data at hand are explored
10.5.6. Follow theoretically and empirically proven specifications and prototypical (reference) use of employed methods (including the official specific ways the developers of methods have presented in the corresponding primary publications)
10.5.7. Models for individual patients: Use dense time series data, leverage population models that apply to the specific individual (including modeling severe and irrevocable rare or singular outcomes), search for and model abrupt distribution shifts of the individual (including learning and modeling shifts at the population level)

- (b) The **causal sparsity (i.e., density or connectivity) of the causal process** that generates the data. A sparse causal data generating process requires less sample size to be discovered because the sample size required for conditional independence tests (CITs) that are at the core of causal structure discovery algorithms increases exponentially (in unrestricted distributions) to the number of conditioning variables in the CIT (see chapter “Foundations of Causal ML”). This latter number is directly linked to the density of the generating causal graph. After a causal graph has been discovered, causal effects of interventions need be estimated and these estimations also require sample size that grows exponentially to the controlled confounders which are similarly linked to the density of the causal generating process.

Whereas classical statistical power-sample analysis can be applied once a good model or causal structure has been identified (to reject the null predictive model or the estimate causal effects, with high confidence) the sample size required for the predictive model discovery or the causal structure discovery and effect estimation are separate considerations. Indeed it is entirely possible for the sample size required for the former to be larger, equal or smaller than the sample size required for the latter.

10.5.8. Best practice strategies to address these sample size and power design needs for ML/AI model building include:

1. Using sensitivity analysis for results over convenience samples by iteratively reducing available sample size (sub-sampling on a convenience sample). If, for example, by reducing the sample size, models retain their predictivity, this strengthens the empirical argument that the learning curve has reached convergence and additional sample will not increase performance.
2. Use of simulations, ideally with real life data where ground truth models are known or re-simulation (as covered in chapter “Principles of Rigorous Development and of Appraisal of ML and AI Methods and Systems”).
3. Use of domain knowledge (if it exists) about the nature of causal structure underlying the data, or the nature of predictive or causal functions to be learned.
4. Use of network-scientific knowledge about the nature of connectivity of real life networks.
5. Reference to prior robust results in very similar domains to formulate and justify assumptions about a successful analysis.
6. Use of dynamic sampling schemes such as adaptive trial designs, Bayesian posterior updating or active learning-based sampling.

Additional Notes on Strategies and Best Practices for Detection, Analysis and Managing Both OC and UP

- (a) **Label reshuffling tests.** The label reshuffling procedure (as for example employed by R. Simon et al. and elaborated in chapter “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models”) retains the joint distribution of input variables however it decouples (on average) the inputs to the response variable. Thus it creates an “on average” null distribution from which we sample data and build models.
 - By comparing the performance of the best model we found to this null distribution we can test the statistical hypothesis that it is as good as random choice (i.e., not reject the null).
 - Additionally, by looking at the mean of this null distribution we can establish whether the overall analysis protocol biases the error estimates (under the null) and by how much.
- (b) **Reanalysis** (with both original and unbiased or otherwise improved protocols, including single coordinated protocols as needed). This is especially important when one wishes to verify the validity of models produced by third parties. For example, when suspicion exists for selective reporting of analyses, or when suspicion of under fitting exists. It is not possible to conduct definitive “forensic style” re-analyses without having access to the full range of data and modeling protocols used in the original analyses. We caution that the ability to run “black box code” on the same data and reproduce the exactly same results, is inappropriately presented as a top-tier level of confidence by some guidelines (see discussion in chapter “Lessons Learned from Historical Failures, Limitations and Successes of AI/ML in Healthcare and the Health Sciences. Enduring Problems, and the Role of Best Practices”), yet it does not suffice because depending on how the black box operates, the models and performance estimates may be OC, UF, or both.
- (c) **“Safety net” model application measures** for ensuring that a model is not applied to the wrong person or population (see chapters “The Development Process and Lifecycle of Clinical Grade and Other Safety and Performance-Sensitive AI/ML Models” and “Characterizing, Diagnosing and Managing the Risk of Error of ML and AI Models in Clinical and Organizational Application” for details on models’ “knowledge cliff” and managing prediction risk).
- (d) **Model stability considerations.** Variable coefficients or very large variation of output given small change in input variable inputs must be dealt with caution as they may imply OF/OC or UP/UF. However we note (without going into full technical details that would require very substantial space to cover), that it is entirely possible for unstable models, markers, causal edges, coefficients etc. to be meaningful and reproducible because of underlying equivalence classes in the data.

Therefore unstable findings should be examined more deeply, but not discarded outright.

In summary the following best practices allow managing (i.e., preventing, diagnosis and/or correcting) OC and UP:

Best Practice 10.1

Deploy procedures that prevent, diagnose and remedy errors of over confidence in, or over fitting of, models.

10.1.1. Manage model complexity with respect to data generating function complexity and to available sample size using:

1. Regularization
2. Dimensionality reduction
3. Feature selection
4. Bayesian Priors and Bayesian ensembles
5. Algorithm-embedded capacity control
6. Statistical model/data complexity measures
7. Model selection
8. Combination approaches

10.1.2. Characterize and manage statistical uncertainty.

10.1.3. Use unbiased estimators of model performance or correct bias of biased estimates.

10.1.4. Lock models at predefined stages in the modeling process and not allow further tampering with locked models.

10.1.5. Correct multiple statistical hypotheses tests (explicitly or implicitly).

10.1.6. Thoroughly specify and report the entirety of procedures used to obtain models so that independent verification of generalizability is possible.

10.1.7. Conduct iterative or sequential modeling via unbiased protocols.

10.1.8. Use representative datasets, appropriate populations and make generalizability claims from appropriate datasets.

10.1.9. Coordinate analysis over many teams and same data via appropriate unbiased protocols.

10.1.10. Use reproducible, standardized data input steps.

10.1.11. Employ normalization /data transforms that do not require entirety of sample (or confine such within discovery and validation datasets independently).

10.1.12. Prevent learners from learning the wrong patterns via spurious co-occurrence; control structural relations and biased sampling; and incorporate domain knowledge-based review that reveals spurious learning.

10.1.13. Control via nested model selection all (and not just a few) factors that can lead to OC.

10.1.14. If possible, combine modelling of individual patients with population modeling.

10.1.15. Do not over-interpret the generalizability of bespoke hand-created AI models unless sufficient number of validation data sets can be obtained to support such claims. Consider creating computable versions of model hand-crafting modeling when possible.

10.1.16. Use label reshuffling testing for evaluating the overfitting/overconfidence bias of the whole analysis protocol.

10.1.17. Apply with appropriate caution Independent dataset validation and be mindful of dangers of over-interpretation of positive and negative results.

10.1.18. Instead of pursuing strict and exact reproducibility across datasets, study the variant and invariant findings from these datasets.

10.1.19. Whenever possible, use reanalysis (with both original and unbiased or otherwise improved protocols, including single coordinated protocols as needed) when verifying the validity of models produced by third parties.

10.1.20. Use domain knowledge and related face-validity tests by experts to flag potential model errors. The experts themselves may be prone to biases or domain theory may not cover models' new findings so do not over-interpret experts' objections.

10.1.21. Apply "safety net" measures for ensuring that a model is not applied to the wrong person or population.

10.1.22. Examine stability of models, parameters and other findings and examine more deeply unstable findings. Be aware that it is possible for unstable findings and models to be perfectly valid.

Best Practice 10.2

Deploy procedures that prevent, diagnose and remedy errors of model under-performance or underfitting.

10.5.1. To maximize predictivity, deploy and explore all relevant learning method families in model selection.

10.5.2. To maximize predictivity and generalizability, deploy and explore all relevant data preparation steps to the domain and task at hand.

10.5.3. To maximize predictivity, systematically and sufficiently explore the hyper parameter space.

10.5.4. Anticipate several preliminary and refinement modeling stages and incorporate them into sequential nested designs to avoid overfitting.

10.5.5. Inform analyses by methods literature so that best known methods for task and data at hand are always explored along with novel methods.

10.5.6. Follow theoretically and empirically proven specifications of reference prototypical or official use of employed methods.

10.5.7. In models for individual patients: use dense time series data, leverage population models, search for and model abrupt distribution shifts of the individual (including learning and modeling shifts at the population level).

10.5.8. Conduct power sample analysis and more generally characterize the effects of sample size on modeling. In the absence of knowledge of learning curves, use:

1. Dynamic sampling schemes whenever appropriate,
2. Sensitivity analysis for results over convenience samples by iteratively reducing available sample size (sub-sampling on a convenience sample),
3. Simulations,
4. Domain knowledge,
5. Network-scientific knowledge,
6. Reference to prior robust results in very similar domains,
7. Dynamic sampling schemes.

Key Concepts Discussed in Chapter “Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI”

Training data error of a model

True generalization error of a model

Estimated generalization error of a model

Overfitting a model to data

Overfitting ML/AI method, system, stack, or protocol

Underfitting a model to data

Underfitting ML/AI method, system, stack, or protocol

Over confidence in a model

Under confidence in a model

Over performance of a model

Under performance of a model

Confidence Interval (CI) of a point estimate P of the performance of a model

Predictive Interval (PI) of a point estimate P of the performance of a model.

Analysis creep

Sequential, iterative and multi-team analyses

Pitfalls Discussed in Chapter “Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI”

Pitfall 10.1.: Producing models in which we have over-confidence

10.1.1. Models are allowed to have inappropriately large complexity with respect to data generating function complexity and to available sample size.

10.1.2. Ignoring statistical uncertainty of point estimates of performance.

10.1.3. Using biased estimators or introducing bias in unbiased ones.

10.1.4. Not correcting multiple statistical hypotheses tests.

10.1.5. Selectively reporting strongest models/results.

10.1.6. Conducting uncontrolled iterative modeling and succumbing to “analysis creep”.

10.1.7. Using non-representative datasets, unusual populations, and making strong claims from too few or too easy datasets.

10.1.8. Not coordinating analysis over many teams and same data via appropriate unbiased protocols designed for collaborative work or competitions.

10.1.9. Using hard-to-reproduce, non-standardized data input steps.

10.1.10. Employing normalization /data transforms that require entirety of sample.

10.1.11. Allowing learners to learn the wrong patterns via spurious co-occurrence, uncontrolled structural relations and biased sampling; and ignoring domain knowledge that reveals the above.

10.1.12. Controlling only some of the factors that can lead to OC.

10.1.13. Inappropriate modelling for individual patients and over interpretation of their generalizability.

10.1.14. Insufficient studies of scalability and generalizability in bespoke hand-created AI models.

Pitfall 10.2.: Producing models that are under performing.

10.2.1. Not deploying the right model family in model selection; not exploring enough method families in model selection.

10.2.2. Insufficient data preparation.

10.2.3. Insufficient exploration of the hyper parameter space during model selection.

10.2.4. 1-stage modeling.

10.2.5. Ignoring best known methods for task and data at hand (either as baseline comparators or starting point).

10.2.5. Ignoring official specifications and prototypical (reference) use of employed methods.

10.2.6. Models for individual patients: lack of dense time series data, failure to leverage population models that apply to the specific individual (including ignoring or under-modeling severe and irrevocable rare or singular outcomes), not addressing well abrupt distribution shifts of the individual (whereas same shifts may be learnable at the population level).

10.2.7. Failing to obtain power sample analysis and more generally control effects of sample size on modeling.

Best Practices Discussed in Chapter “Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI”

Best Practice 10.1: Deploy procedures that prevent, diagnose and remedy errors of over confidence in, or over fitting of, models.

10.1.1. Manage model complexity with respect to data generating function complexity and to available sample size using:

1. Regularization,
2. Dimensionality reduction,
3. Feature selection,
4. Bayesian Priors and Bayesian ensembles,
5. Algorithm-embedded capacity control,
6. Statistical model/data complexity measures,
7. Model selection,
8. Combination approaches.

10.1.2 Characterize and manage statistical uncertainty.

10.1.3. Use unbiased estimators of model performance or correct bias of biased estimates.

10.1.4. Lock models at predefined stages in the modeling process and not allow further tampering with locked models.

10.1.5. Correct multiple statistical hypotheses tests (explicitly or implicitly).

10.1.6. Thoroughly specify and report the entirety of procedures used to obtain models so that independent verification of generalizability is possible.

10.1.7. Conduct iterative or sequential modeling via unbiased protocols.

10.1.8. Use representative datasets, appropriate populations, and make generalizability claims from appropriate datasets.

10.1.9. Coordinate analysis over many teams and same data via appropriate unbiased protocols.

10.1.10. Use reproducible, standardized data input steps.

10.1.11. Employ normalization /data transforms that do not require entirety of sample (or confine such within discovery and validation datasets independently).

10.1.12. Prevent learners from learning the wrong patterns via spurious co-occurrence; control structural relations and biased sampling; and incorporate domain knowledge-based review that reveals spurious learning.

10.1.13. Control via nested model selection all (and not just a few) factors that can lead to OC.

10.1.14. Carefully combine modelling of individual patients with population modeling.

10.1.15. Do not over-interpret the generalizability of bespoke hand-created AI models unless sufficient validation data can be obtained to support such claims. Consider creating computable versions of model hand-crafting modeling when possible.

10.1.16. Use label reshuffling testing for evaluating the overfitting bias of the whole analysis protocol.

10.1.17. Apply with appropriate caution independent dataset validation and be mindful of dangers of over interpretation of positive and negative results.

10.1.18. Instead of pursuing strict and exact reproducibility across datasets, study the variant and invariant findings across these datasets.

10.1.19. Whenever possible, use reanalysis (with both original and unbiased or otherwise improved protocols, including single coordinated protocols as needed) when verifying the validity of models produced by third parties.

10.1.20. Use domain knowledge and related face-validity tests by experts to flag potential model errors. The expert themselves may be prone to biases or domain theory may not cover models' new findings so do not over-interpret experts' objections.

10.1.21. Apply "safety net" measures for ensuring that a model is not applied to the wrong person or population.

10.1.22. Examine stability of models, parameters and other findings and examine more deeply unstable findings. Be aware that it is possible for unstable models to be perfectly valid.

Best Practice 10.2: Deploy procedures that prevent, diagnose and remedy errors of model under performance or underfitting.

10.5.1. Deploy and explore all relevant learning method families in model selection.

10.5.2. Deploy and explore all relevant data preparation steps to the domain and task at hand.

10.5.3. Systematically and sufficiently explore the hyper parameter space.

10.5.4. Anticipate several preliminary and refinement modeling stages and incorporate in sequential nested designs to avoid overfitting.

10.5.5. Inform analyses by methods literature so that best known methods for task and data at hand are always explored along with novel methods.

10.5.6. Follow theoretically and empirically proven specifications of reference prototypical or official use of employed methods.

10.5.7. In models for individual patients: use dense time series data, leverage population models, search for and model abrupt distribution shifts of the individual (including learning and modeling shifts at the population level).

10.5.8. Conduct power sample analysis and more generally characterize the effects of sample size on modeling. Use:

1. Dynamic sampling schemes whenever appropriate,
2. Sensitivity analysis for results over convenience samples by iteratively reducing available sample size (sub-sampling on a convenience sample),
3. Simulations,
4. Domain knowledge,
5. Network-scientific knowledge,
6. Reference to prior robust results in very similar domains,
7. Dynamic sampling schemes.

Classroom Assignments and Discussion Topics

Chapter “Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI”

- 1. Consider Rich Simon’s experiment with the following modification: the feature selection method used is based on univariate association using a Hockberg-Benjamini control of false positive rate at 10%. This means that the features will be ranked by p-value of the association with the response variable and thresholded so that no more than 10% of selected features will be false positive correlates of the response variable.
 - (a) How many features will be selected if no feature has true signal for the response?
 - (b) What will be the error estimation bias of complete, incomplete and no cv schemes?
 - (c) Based on the above, does the Simon et al. conclusions hold regardless of the feature selection method?
 - (d) How would you modify the Simon et al. guidance?
- 2. Is it possible that error of a classifier is 0 in the TR data, true optimal generalization error is >0 and this classifier is optimal? In other words, is it possible for a model to be optimal yet overfitted?
- 3. Consider the following scenario describing two model selection procedures MS1, MS2 and MS3 each considering and selecting different sets of models with estimated and true generalization errors as described in the table. For simplicity assume no other models can be fitted in this setting.

	Model 1	Model 2	Model 3	Model 4
TR accuracy	0.95	0.90	0.75	0.65
Estimated generalization accuracy by CV	0.80	0.88	0.78	0.70
Estimated generalization accuracy by Estimator X	0.9	0.96	0.92	0.80
True generalization accuracy	0.80	0.88	0.76	0.73
MS1 selects model	Yes	No	Yes	Yes
MS2 selects model	Yes	Yes	Yes	Yes
MS3 considers model	No	Yes	No	Yes

- (a) What are you conclusions about OP, OC, OF, UF and UP of models 1 to 4?
 - (b) What is your assessment of the bias of the generalization accuracy estimator X used here?
 - (c) Why CV does not exactly match the true generalization error, although it is unbiased?
 - (d) Which models are selected by each of MS1 to MS3? How would you characterize these model selectors?
- 4. [ADVANCED] The label reshuffling procedure tests whether a model is statistically significantly different than a model without signal and simultaneously whether the overall modeling protocol has a propensity for producing over

- confident estimates. This holds under the null hypothesis (i.e. there is no signal in the data).
- (a) The reshuffling takes place only in the response variable labels. Explain why we DO NOT randomize all variables (i.e., both inputs and outputs).
 - (b) Bonus/research topic open problem: can you think of possible procedures that could test against alternative hypotheses (i.e., against a user-postulated non-zero signal)?
5. Consider years 2020–2021 of the COVID epidemic whereas many factors were constantly changing.
- (a) What are some key factors that were changing?
 - (b) What challenges of the OF/UP/OC varieties does a situation like this creates for various types of AI/ML decision models? Consider ICU admission decision models as an example.
6. Describe, by example or more general analysis, how a researcher can produce clustering omics data so that a published cluster model exhibits good diagnostic accuracy even though no such signal exists in the data.
7. Comment on the following position: “whenever the true signal in the data is very high, it is more difficult to produce models with serious overconfidence errors; conversely as true signal approaches zero, the magnitude of possible OC error increases”. What are underlying assumptions in the above thesis?
8. Consider the following (idealized and simplified) modeling situation. A data scientist is tasked by her manager to create a predictive model. She decides to use nested hold out (equivalent to NNCV with one fold) as follows: the total data is randomly split in mutually exclusive datasets TRTR TRTE and TE. She ensures that the prior of the binary response variable is the same in all three datasets. She considers 2 possible values for hyper-parameter H of ML algorithm A and estimates accuracy (0/1 error) as follows:

Algorithm A				
Accuracy in TRTE for H = 1	Accuracy in TRTE for H = 2	Best value of H	Accuracy in TE of model with best value of H	Finally reported accuracy of best model
0.80	0.90	2	0.85	0.85

She presents the results to the project manager who suggests that a second algorithm B is used because it may increase accuracy. The results this time look like this:

Algorithm B				
Accuracy in TRTE for H = 1	Accuracy in TRTE for H = 2	Best value of H	Accuracy in TE of model with best value of H	Finally reported accuracy of best model
0.85	0.80	1	0.80	0.80

She presents the results to the project manager who still is not satisfied with the result and brings in a consultant who suggests that a third algorithm C is used. The results this time look like this:

Algorithm C				
Accuracy in TRTE for H = 1	Accuracy in TRTE for H = 2	Best value of H	Accuracy in TE of model with best value of H	Finally reported accuracy of best model
0.75	0.80	2	0.90	0.90

Based on the above the manager and the consultant conclude that the model produced by algorithm C is the best, it has generalization accuracy 0.90, and should be deployed.

The data scientist (who recently read chapter “Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI”) is not convinced however because she now suspects that an “analysis creep” situation has occurred leading to overconfidence in a model. She decides to conduct a nested holdout based analysis, this time analyzing all 3 algorithms simultaneously in the nested part (inner loop) of the cv design. The following table presents her results:

Algorithm A	Accuracy in TRTE for H = 1	Accuracy in TRTE for H = 2	Best algorithm/best value of H: Algorithm A, H = 2	Accuracy in TE of model with best algorithm/best value of H 0.85	Finally reported estimated generalization accuracy of best model 0.85
Algorithm B	Accuracy in TRTR for H = 1	Accuracy in TRTE for H = 2			
	0.85	0.80			
Algorithm C	Accuracy in TRTR for H = 1	Accuracy in TRTE for H = 2			
	0.75	0.80			

The manager and the consultant are perplexed.

Assume the role of the data scientist and write a short report explaining how an OC error occurred in the first round of analyses and why the second analysis is unbiased. For simplicity, ignore the need to conduct tests of statistical difference between point estimates and interpret nominal accuracy point estimates as true ones.

9. We saw that combining capacity control mechanisms provides augmented protections against overfitting/excessive capacity.
- (a) Describe an existing protocol of your choice (or one that you construct) that combines 4 or more ways to control excessive model capacity.

(b) Bonus question [ADVANCED]: is it possible for such combinations to have negative effects on ability to control the capacity of the models produced?

10. [ADVANCED]
 - (a) Show that in order to calculate the true positives rate (aka PPV) or a model's decisions we need to know the prior of the distribution of the response variable.
 - (b) Show that an active learning sampling design alone does not allow the analyst to know this prior whereas a random sampling design does.
 - (c) What are the implications therefore of an active learning design for controlling the PPV?
11. Is it ok to build under performing models in the context of exploratory research? Present a few situations where it might be a good idea and some where it is a bad idea.
12. "Ocam's Razor" is an epistemological principle that says that between two models that explain the data equally well, the simpler one is more likely to be true. How does this principle relate to the BVDE? Can you think of a counter example (HINT: consider causal modeling).
13. (a) Describe how PCA can lead to OC errors.
 (b) Consider R. Simon's experiment where instead of feature selection the analyst use a PC-mapping of the input data such that correlation with the response variable is maximized. Is this subject to the same bias that Simon et al. described?
 (c) How would you conduct unbiased error estimation using NNFCV whereas the data is PCA-transformed?
14. [ADVANCED] Wolpert uses NFLT and OTSE to argue in [19] that cross validation is not better as a model selection strategy than *doing the exact opposite* (i.e., choose the model with highest error in the test data), which he coins "anti-cross validation". If he is right, then what that Chapter "Overfitting, Underfitting and General Model Overconfidence and Under-Performance Pitfalls and Best Practices in Machine Learning and AI" teaches as best practices for model selection and error estimation is only a heuristic strategy that may fail in as many situations as the ones that it will succeed. Refute these claims.
 HINT: focus your arguments either around the misalignment of OTSE with real-life modeling objectives/performance metrics, or alternatively/additionally with the misalignment of these objectives with averaging over all distributions rather the distribution in hand.

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